S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

FinRisk Intelligence

Turning unstructured financial signals into explainable risk intelligence and portfolio-level stress analysis.

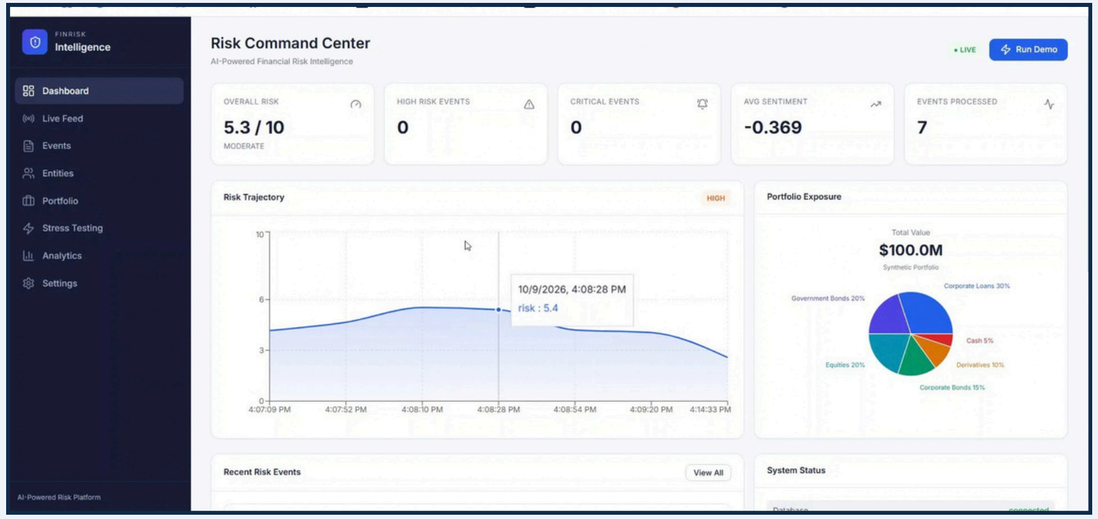
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Individual submission

▶ **DEMO VIDEO:** https://youtu.be/t5AkG_-akgY

GitHub: github.com/PulkitAg13/VIT_Bhopal_University-Pulkit_Agrawal-Hackathon_



Risk Command Center: synthetic demo data, model-generated scores

1	**What is being said?** Sentiment (FinBERT)
2	**What event is happening?** Event class + entities
3	**How important is it?** Explainable impact score
4	**What could it mean for a portfolio?** Exposure + stress test

Problem & Approach

THE PROBLEM: INFORMATION OVERLOAD

Not a lack of information, but a lack of prioritization.

- Financial news
- Social / media signals
- Market information
- Company announcements

EVERY ANALYST NEEDS TO KNOW

- ? What happened?
- ? Which entity is affected?
- ? What type of event?
- ? How severe is it?
- ? New or a duplicate?
- ? What portfolio exposure?

OUR APPROACH: AN END-TO-END PIPELINE

1. **Financial information**
   Yahoo Finance RSS, public datasets
2. **AI understanding**
   FinBERT, BART-MNLI, BERT-NER, MiniLM
3. **Risk intelligence**
   Event, impact, confidence, novelty
4. **Portfolio impact**
   Entity exposure on a synthetic $100M book
5. **Decision support**
   Stress scenarios and analytics

Not just sentiment analysis: from "what is being said" to "what could this mean for a portfolio?"

System Design & Data Flow

DATA SOURCES	INGESTION	NLP ENGINE	RISK INTELLIGENCE	PORTFOLIO & STRESS	COMMAND CENTER
Yahoo Finance RSS Twitter Financial News (historical replay) Public datasets Manual / demo input	FastAPI backend Text preprocessing Ticker selection	FinBERT sentiment BART-MNLI event class BERT-NER entities MiniLM embeddings	Dedup + novelty Clustering, corroboration Explainable impact score (1-10)	Entity to exposure Synthetic $100M, 18 positions Configurable stress scenarios	Live Feed, Events Entities, Portfolio Stress Testing Analytics

REAL-TIME PATH

Docker Compose runs PostgreSQL, Redis, backend and frontend.

PostgreSQL stores events › **Redis Pub/Sub** broadcasts › **WebSocket** pushes live › **React UI** updates instantly

STRUCTURED RISK EVENT Source · Text · Timestamp · Entity · Sentiment · Event class · Impact score · Risk level · Confidence

Implementation Highlights

01 Finance-specific sentiment

ProsusAI/FinBERT

Finance-oriented model, not generic sentiment. Scores from -1 (negative) to +1 (positive).

02 Zero-shot event classification

facebook/bart-large-mnli

Classifies against a configurable taxonomy (geopolitical, macro, credit, M&A, earnings, supply chain) with no fixed classifier.

03 Entity intelligence

dslim/bert-base-NER

Transformer NER plus canonical resolution links each event to companies such as Apple or Amazon.

04 Novelty & clustering

all-MiniLM-L6-v2

Semantic similarity: repeated or syndicated stories are not counted as new events, so alerts stay clean.

05 Explainable impact score

Multi-factor risk fusion

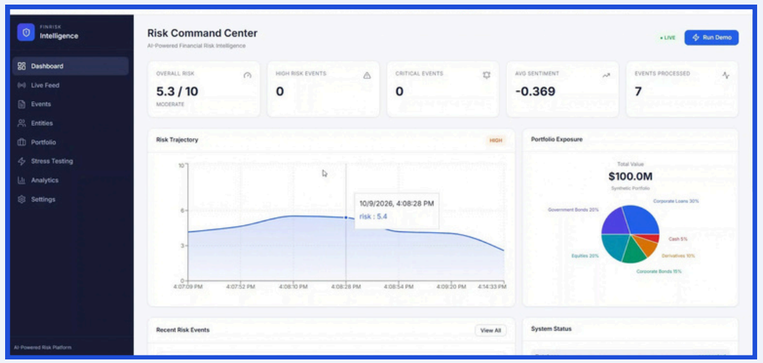
Sentiment, event type, entity, novelty, corroboration and recency give a 1-10 score, a risk level and "Why this score?".

06 Real-time full stack

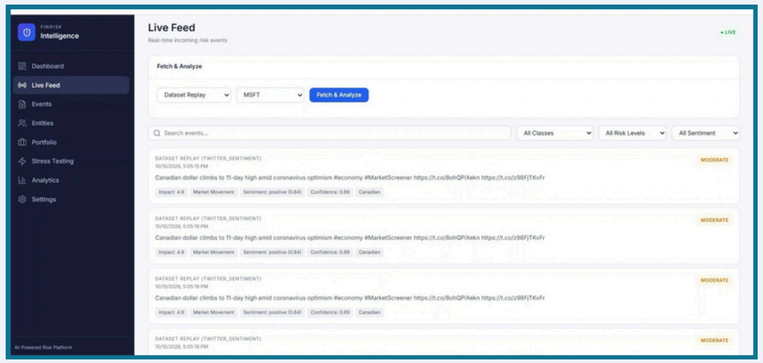
FastAPI · PostgreSQL · Redis · React

WebSocket pushes new events live. Docker Compose, automated tests, public MIT-licensed repo.

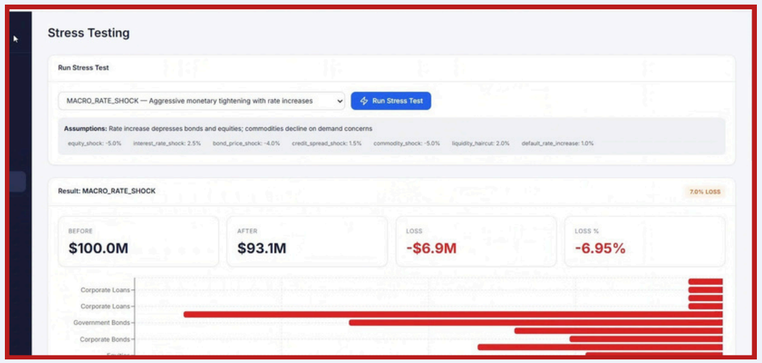
Key Results



DASHBOARD | MODEL-GENERATED



LIVE FEED | PUBLIC DATASET REPLAY



STRESS TEST | SIMULATED

5.3 / 10

Overall risk, demo state

$100M

Synthetic portfolio, 18 positions, 6 asset classes

-6.95%

Simulated macro rate shock

~$6.9M

Simulated loss, synthetic portfolio

EFFICIENCY VS. A NAIVE APPROACH

Duplicates Every copy is an alert → **Novelty-checked, clustered**

Sentiment Generic model → **Finance-oriented FinBERT**

Event type Fixed classifier → **Zero-shot, configurable**

Score Opaque number → **Multi-factor, explained**

Outcome Text label only → **Portfolio stress impact**

Domain Impact

01

FASTER DECISIONS

Relevant events surface earlier from the noise.

Impact score + risk level

02

LESS ALERT FATIGUE

Duplicates and corroborating stories are consolidated.

Dedup + clustering

03

EXPLAINABLE & AUDITABLE

Every score has a reason a risk team can review.

"Why this score?"

04

FORWARD-LOOKING RISK

Events connect to portfolio exposure and stress.

Exposure + stress test

SIGNAL → PRIORITY → EXPOSURE → SCENARIO → DECISION

Public + synthetic data only

Scores are model outputs, not market truth

Stress results are simulations, not forecasts

Limitations & Next Steps

ASSUMPTIONS & LIMITATIONS

- Portfolio is synthetic ($100M, 18 positions), so it shows the mechanism, not real exposure
- Stress results are simulations, not forecasts of real losses
- Twitter component is a historical dataset replay, not live streaming
- Impact, confidence and risk scores are analytical model outputs
- Public data only; no proprietary S&P Global or Crisil data

NEXT EVOLUTION

- Finance-tuned NER + entity linking
- Faster, distilled event classifiers
- Position / counterparty-level mapping
- Back-testing and score calibration
- Filings, transcripts and alert routing

From unstructured financial signals to explainable portfolio risk.

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